

SHOWA STATION, Antarctica

WMO: 895320

Lat: 69.005S

Lon: 39.581E

Elev: 18

StdP: 101.10

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-33.4	-30.8	-37.5	0.1	-32.3	-35.6	0.1	-29.9	30.3	-9.8	27.7	-10.2	2.4	200	1.243

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	3.5	0.2	2.6	-0.4	1.7	-0.8	0.6	2.7	0.0	1.9	-0.5	1.3	3.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
-0.9	3.5	0.4	-1.7	3.3	0.1	-2.5	3.1	-0.1	10.8	2.9	9.8	2.0	8.9	1.4	3.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.1	23.2	20.0	-35.7	5.6	3.1	1.1	-37.9	6.4	-39.7	7.0	-41.5	7.6	-43.7	8.4		
(5)			-35.8	1.7	3.1	0.8	-38.0	2.3	-39.8	2.7	-41.6	3.2	-43.8	3.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.6	-1.1	-3.0	-6.9	-10.4	-13.8	-15.3	-17.5	-18.7	-18.4	-13.5	-6.5	-1.7
	DBStd	7.76	1.84	2.42	3.33	4.40	5.21	5.57	6.24	6.08	5.95	4.72	3.66	1.93
	HDD10.0	7521	344	364	524	612	736	758	853	890	852	729	495	363
	HDD18.3	10562	603	597	782	862	994	1008	1111	1148	1102	988	745	621
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.7	4.9	7.0	7.7	8.5	7.7	7.4	7.1	6.4	6.2	6.5	6.1	5.4
Precipitation	PrecAvg	416	35	46	49	46	38	32	32	29	21	30	30	27
	PrecMax	538	67	81	70	74	73	64	79	53	46	60	64	60
	PrecMin	318	14	26	16	13	13	8	12	11	5	12	2	4
	PrecStd	52	15	15	12	13	17	15	14	11	10	11	15	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	3.6	-0.1	-1.3	-3.2	-3.3	-5.0	-5.4	-5.1	-2.8	2.0	4.6
		MCWB	1.1	0.1	-1.5	-1.6	-3.9	-3.9	-5.6	-6.2	-5.9	-3.7	-1.3	1.0
	2%	DB	3.8	1.9	-0.9	-2.9	-4.4	-4.5	-6.4	-7.2	-6.8	-4.0	0.6	3.2
		MCWB	0.3	-0.7	-2.2	-3.6	-5.1	-5.2	-7.1	-7.9	-7.6	-4.8	-2.0	0.1
	5%	DB	2.8	1.0	-1.8	-3.9	-5.5	-5.8	-7.8	-8.6	-8.6	-5.5	-0.6	2.4
		MCWB	-0.2	-0.9	-3.1	-4.7	-6.2	-6.5	-8.5	-9.3	-9.3	-6.4	-2.9	-0.5
	10%	DB	2.0	0.3	-2.6	-4.8	-6.9	-7.2	-9.0	-10.1	-10.2	-6.7	-1.6	1.5
		MCWB	-0.7	-1.4	-3.9	-5.7	-7.7	-7.9	-9.8	-10.8	-11.1	-7.7	-3.7	-1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	0.9	-1.0	-1.6	-3.5	-3.7	-5.4	-5.8	-5.7	-3.3	-0.8	1.7
		MCDB	4.4	2.5	-0.5	-1.4	-3.3	-3.4	-5.2	-5.4	-5.3	-3.1	1.3	4.1
	2%	WB	0.7	0.1	-1.9	-3.4	-4.9	-5.1	-7.0	-7.8	-7.5	-4.8	-1.8	0.5
		MCDB	3.0	1.1	-1.2	-3.1	-4.6	-4.7	-6.4	-7.3	-6.8	-4.2	0.3	2.7
	5%	WB	0.1	-0.5	-2.9	-4.5	-6.2	-6.4	-8.4	-9.2	-9.2	-6.3	-2.7	-0.3
		MCDB	2.3	0.7	-1.9	-4.1	-5.5	-5.9	-7.8	-8.6	-8.6	-5.6	-0.9	1.8
	10%	WB	-0.4	-1.1	-3.8	-5.6	-7.6	-7.9	-9.8	-10.8	-11.1	-7.6	-3.5	-0.8
		MCDB	1.6	0.2	-2.8	-4.9	-6.9	-7.2	-9.0	-10.1	-10.3	-6.8	-1.8	1.2
Mean Daily Temperature Range	5% DB	MDBR	4.3	3.9	4.1	4.5	4.9	5.5	5.6	6.1	5.9	5.3	5.3	4.4
		MCDBR	5.2	3.9	3.4	3.4	4.1	5.4	5.6	6.3	6.3	5.1	5.7	5.3
		MCWBR	3.6	3.1	3.5	3.7	4.1	5.3	5.6	6.1	6.1	4.8	4.7	3.9
	5% WB	MCDBR	4.5	3.2	3.5	3.5	3.9	5.3	5.6	6.3	6.3	4.8	5.0	4.7
		MCWBR	3.3	2.8	3.7	3.8	4.0	5.4	5.7	6.2	6.3	4.6	4.3	3.6
Clear-Sky Solar Irradiance	taub	0.257	0.252	0.230	0.185	0.129	N/A	0.128	0.168	0.201	0.222	0.238	0.257	
	taud	2.267	2.292	2.312	2.138	1.911	N/A	1.892	2.093	2.251	2.268	2.285	2.254	
	Ebn at Noon	991	934	846	617	108	0	81	633	872	968	1011	1008	
	Edh at Noon	112	93	67	36	4	0	3	35	68	94	109	118	
All-Sky Solar Radiation	RadAvg	7.80	4.99	2.45	0.65	0.06	0.00	0.01	0.36	1.82	4.25	7.25	8.89	
	RadStd	0.36	0.28	0.16	0.05	0.01	0.00	0.00	0.02	0.09	0.16	0.28	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page